

Introduction

Nanotechnology is the field of study concerned with **understanding and manipulating** matter at the **nanoscale**, which is found at **~1-100 nanometers**. For reference, a nanometer is about a billionth (1/1,000,000,000) of a meter. In the field of **personal protective equipment** (mainly in the equipment of **firefighters**), nanotechnology has been used increasingly over the years and can be credited with many of the recent era **improvements to firefighter turnout gear**. Turnout gear is responsible for protecting all parts of firefighter's bodies from **heat and hazardous materials**, especially their **hands**, due to the how **close** they often are to the hazardous materials.

Characteristic	Inpatient stays		ED visits	
	n	%	n	%
Burn site				
Total burn-related stays/visits ^a	53,220	100.0	409,183	100.0
Eye, face, head, or neck	14,015	26.3	75,451	18.4
Trunk	17,515	26.3	75,451	18.4
Upper limb except wrist and hand	16,605	31.2	90,095	22.0
Wrist and hand	14,995	28.2	157,497	38.5
Lower limb	20,500	38.5	94,906	23.2
Multiple specific sites, internal organs	1,775	3.3	8,035	2.0
Unspecified site	400	0.8	4,720	1.2
Late effects of burn	4,145	7.8	2,413	0.6

Figure 1. Data from 2013 hospital study tracking burn related injuries (McDermott, 2016).

One option for increasing the strength of the gear to **prevent heat penetration** is using **silicon dioxide (SiO₂) nanopowder**, or nanosilica, to create a **coating** on the **exterior surface** of the **glove**. Due to its high **specific heat capacity**, **strong hydrophobic properties**, and **thermal stability**, nanosilica is an excellent and industry tested material for positively increasing the **efficiency** and **durability** of firefighter turnout.



Figure 2. Silicon dioxide (SiO₂) nanopowder / nanosilica (Davey, 2024).



Figure 3. Industry standard firefighting gloves, SOLAS glove model (DATREX, 2025).

Purpose

For a group built on a philosophy of “Everyone Goes Home,” where they run into **hazardous situations** to protect their communities while also having no regard for their own safety, firefighters **deserve protection** from their turnout gear in the same manner. Advancements have been made consistently in the field for most pieces of equipment **except for the gloves**. This can be fixed through the use of a **nano-coating** on the **exterior surface**. With **heat testing**, it can be determined whether or not the nanosilica can function as an effective option if it **decreases the heat permeating through** to the hand by a **measurable amount**. Along with this, the results from this project can be used **comparatively** to find possible new options for **hand protection**.

Improving the Flame-Retardant Qualities of a Glove Through Nanotech Enhancements for Upper Extremity Personal Protective Equipment

Methods

Step 1: Creating the Resin and Nanosilica Solution



Figure 4. Solution of resin, hardener, and nanosilica in beaker.

Combined 2 oz of Part A (resin) with 1 oz of Part B (hardener) in a container to maintain a ratio of 2:1. Stirred the solution with a stir rod until the overall texture of the mixture is smooth. After that, to keep a 1:5 ratio for the nanosilica to the solution, ~0.6 oz of nanosilica (SiO₂) was added to the mixture. Mixed the solution for approximately 5 minutes until it was fully combined.

Step 2: Coating the Glove with the Solution



Figure 5. Coating the prototype glove using the created solution.

With the appropriately created resin/nanosilica solution, a paintbrush was used to carefully coat the exterior surfaces of only one of the gloves to keep a control glove to measure against. The glove received one cohesive coat with special care around the finger joints and glove opening to maintain as much dexterity and structure as possible. The glove was left to cure for 3 days after the coat was applied.

Step 3: Heat Testing the Coated Glove



Figure 6. Heat testing setup inside of the fume hood.

After the glove was done curing, changes in mass, dexterity, and overall structure were observed and noted. Then, a comparative study was performed between the coated and non-coated glove using a heat gun on two different temperature settings and a thermocouple to measure the heat that permeated through the glove.

Results

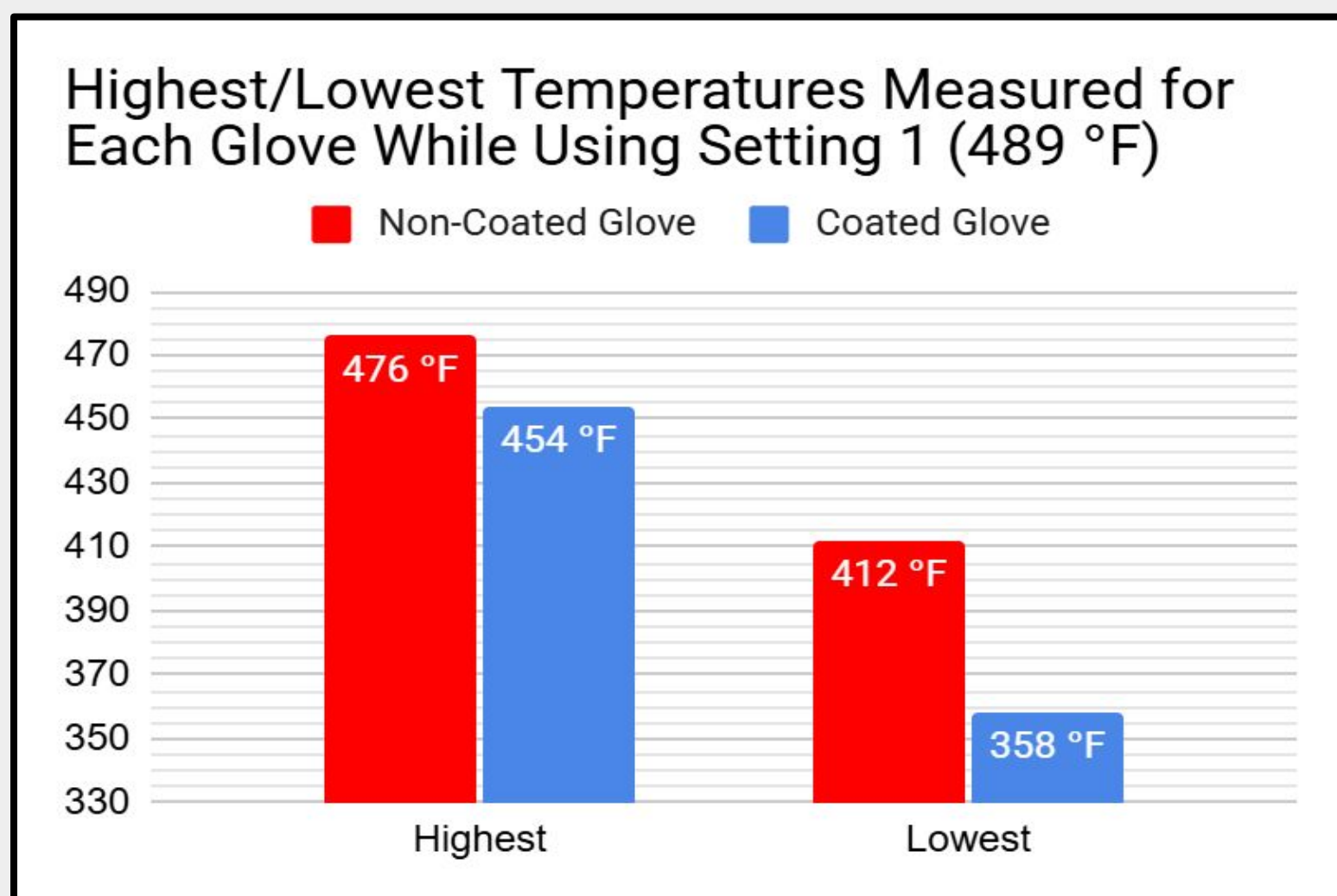


Figure 7. The first test used the heat gun on its low setting which heated to ~489 °F. The graph illustrates a comparison between the highest and lowest temperatures observed by the thermocouples recorded temperatures during each of the tests.

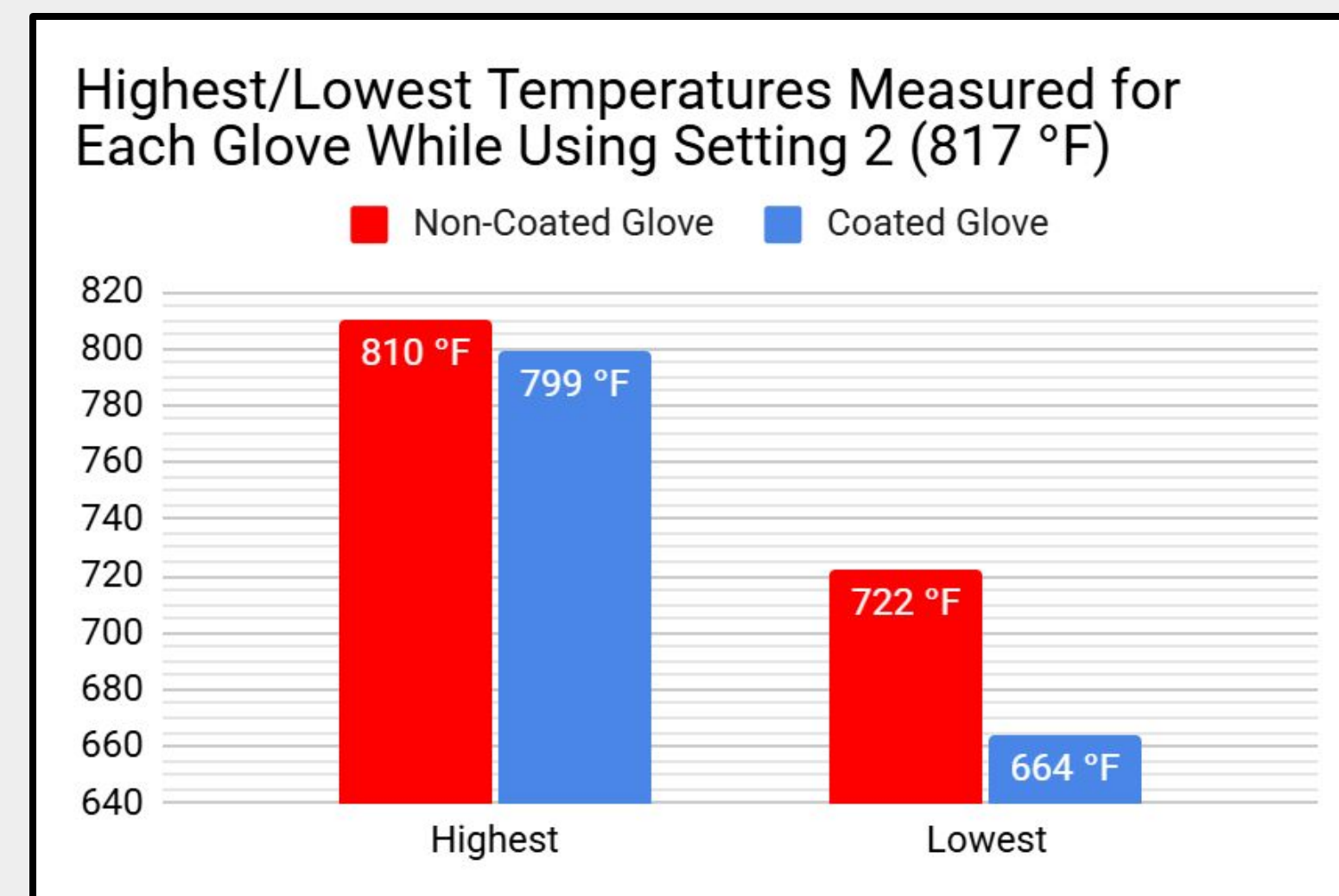


Figure 8. The second test used the heat gun on its high setting which heated to ~817 °F. The graph illustrates a comparison between the highest and lowest temperatures observed by the thermocouples recorded temperatures during each of the tests.

Conclusions

By comparing the data collected from each of the two tests, it can be concluded that the glove overall was **successful in decreasing the heat moving** through the **exterior surface** affecting the **internal temperature**. Both tests returned the same results that the **coated glove was able to prevent more heat** than the **non-coated glove**. Confirming this numerically, the coated glove **prevented 11% more heat** on the low setting and **7.1% more heat** on the high setting. Based on this alone, the **project was a success**. However, the **original goal** was to **decrease heat** while **maintaining the same structure and dexterity**. The first part was done, but the coated glove became **extremely rough** and the **flexibility** was **limited heavily**. This likely occurred due to **excess solution** being applied to the **fabric parts of the glove**. If the **application process** was **performed slower**, a glove that still had improved heat prevention while **maintaining flexibility** could be made.



Figure 9. Non-coated glove undergoing the heat testing procedure that created the data detailed above.

Extensions

Although, the **final product** didn't perfectly align with the **original goals**, the project as a whole was still **relatively successful** and showed the **possible effectiveness** of the procedure. The nanomaterial **application** did have a **real impact** in **preventing heat from permeating through the glove's surface**. The relative success of this project can be a point of reference for **improving firefighting gloves** and **personal protective equipment** as a whole. While **issues** did come up such as the **loss of dexterity**, the project can still be a example of the **possible downsides** to **nano-applications** on **fabric-based materials**.

References

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